

9월 16일 미팅 자료 김성현

대상 소프트웨어 : ZOOKEEPER

비고

직접적인 크래시가 아닌

RaceCondition,

로직 에러,

deadlock,

hang(기다림) 등의 에러를 조사하였음

이슈번호 1069

Calling shutdown() on a QuorumPeer too quickly can lead to a corrupt log

QuorumPeer1이 종료 요청 but, 종료가 보장되지 않았음

뒤늦게 내부 스레드가 동작

트랜잭션 로그가 반만 만들어짐 (헤더가 매직넘버가 아닌 0)

이후 재기동했을 때 불완전한 트랜잭션 로그를 읽고 기동을 하지 않음

```
2011-05-18 22:42:29,823 10467 [pool-1-thread-2] FATAL org.apache.zookeeper.server.quorum.QuorumPeer - Unable to load database on disk
java.io.IOException: Transaction log: /var/cloudnet/data/zookeeper/version-2/log.700000001 has invalid magic number 0 != 1514884167
    at org.apache.zookeeper.server.persistence.FileTxnLog$FileTxnIterator.inStreamCreated(FileTxnLog.java:510)
    at org.apache.zookeeper.server.persistence.FileTxnLog$FileTxnIterator.createInputArchive(FileTxnLog.java:527)
    at org.apache.zookeeper.server.persistence.FileTxnLog$FileTxnIterator.goToNextLog(FileTxnLog.java:493)
    at org.apache.zookeeper.server.persistence.FileTxnLog$FileTxnIterator.next(FileTxnLog.java:576)
    at org.apache.zookeeper.server.persistence.FileTxnLog$FileTxnIterator.init(FileTxnLog.java:479)
    at org.apache.zookeeper.server.persistence.FileTxnLog$FileTxnIterator.<init>(FileTxnLog.java:454)
    at org.apache.zookeeper.server.persistence.FileTxnLog.read(FileTxnLog.java:325)
    at org.apache.zookeeper.server.persistence.FileTxnSnapLog.restore(FileTxnSnapLog.java:126)
    at org.apache.zookeeper.server.ZKDatabase.loadDataBase(ZKDatabase.java:222)
    at org.apache.zookeeper.server.quorum.QuorumPeer.start(QuorumPeer.java:398)
```

...

```
2011-05-18 22:42:29,823 10467 [pool-1-thread-2] ERROR com.nicira.onix.zookeeper.Zookeeper - Unexpected exception
java.lang.RuntimeException: Unable to run quorum server
    at org.apache.zookeeper.server.quorum.QuorumPeer.start(QuorumPeer.java:401)
    at com.nicira.onix.zookeeper.Zookeeper.StartZookeeper(Zookeeper.java:198)
    at com.nicira.onix.zookeeper.Zookeeper.RestartZookeeper(Zookeeper.java:277)
    at com.nicira.onix.zookeeper.ZKRPCService.setServers(ZKRPC.java:83)
    at com.nicira.onix.zookeeper.Zkrpc$ZKRPCService.callMethod(Zkrpc.java:8198)
    at com.nicira.onix.rpc.RPC$10.run(RPC.java:534)
    at java.util.concurrent.ThreadPoolExecutor$Worker.runTask(ThreadPoolExecutor.java:886)
    at java.util.concurrent.ThreadPoolExecutor$Worker.run(ThreadPoolExecutor.java:908)
    at java.lang.Thread.run(Thread.java:662)
```

```
Caused by: java.io.IOException: Transaction log: /var/cloudnet/data/zookeeper/version-2/log.700000001 has invalid magic number 0 != 1514884167
    at org.apache.zookeeper.server.persistence.FileTxnLog$FileTxnIterator.inStreamCreated(FileTxnLog.java:510)
    at org.apache.zookeeper.server.persistence.FileTxnLog$FileTxnIterator.createInputArchive(FileTxnLog.java:527)
    at org.apache.zookeeper.server.persistence.FileTxnLog$FileTxnIterator.goToNextLog(FileTxnLog.java:493)
    at org.apache.zookeeper.server.persistence.FileTxnLog$FileTxnIterator.next(FileTxnLog.java:576)
    at org.apache.zookeeper.server.persistence.FileTxnLog$FileTxnIterator.init(FileTxnLog.java:479)
    at org.apache.zookeeper.server.persistence.FileTxnLog$FileTxnIterator.<init>(FileTxnLog.java:454)
    at org.apache.zookeeper.server.persistence.FileTxnLog.read(FileTxnLog.java:325)
    at org.apache.zookeeper.server.persistence.FileTxnSnapLog.restore(FileTxnSnapLog.java:126)
    at org.apache.zookeeper.server.ZKDatabase.loadDataBase(ZKDatabase.java:222)
    at org.apache.zookeeper.server.quorum.QuorumPeer.start(QuorumPeer.java:398)
```

이슈번호 1109

dataDir 디스크가 가득 차면
zookeeper가 종료를 시도함

그러나 this.join과 system.exit(11)이 동일 스레드에
서 호출됨

SyncThread
waits for
Shutdown Hook
waits for
SyncThread

이슈번호 1109

순환 대기가 형성됨

디스크가 다 차서 종료되는건 당연.

(나머지가 리더로 선출되면 됩니다.)

이 이슈는 종료 자체가 제대로 안일어나고

순환대기가 일어나버려서 문제가 발생

System.exit를 호출한 SyncThread와 shutdown hook 사이의 순환 대기(deadlock)

```
1 2011-06-21 10:09:59,730 - FATAL [SyncThread:2:SyncRequestProcessor@148] - Severe unrecoverable error, exiting
2 java.io.IOException: No space left on device
3     at java.io.FileOutputStream.writeBytes(Native Method)
4     at java.io.FileOutputStream.write(FileOutputStream.java:260)
5     at java.io.BufferedOutputStream.flushBuffer(BufferedOutputStream.java:65)
6     at java.io.BufferedOutputStream.flush(BufferedOutputStream.java:123)
7     at org.apache.zookeeper.server.persistence.FileTxnLog.commit(FileTxnLog.java:305)
8     at org.apache.zookeeper.server.persistence.FileTxnSnapLog.commit(FileTxnSnapLog.java:324)
9     at org.apache.zookeeper.server.ZKDatabase.commit(ZKDatabase.java:484)
10    at org.apache.zookeeper.server.SyncRequestProcessor.flush(SyncRequestProcessor.java:158)
11    at org.apache.zookeeper.server.SyncRequestProcessor.run(SyncRequestProcessor.java:98)
12 2011-06-21 10:09:59,732 - INFO [Thread-2:QuorumPeer@691] - The Quorum server is going for shutdown
13 2011-06-21 10:09:59,732 - INFO [Thread-2:Leader@393] - Shutdown called
14 java.lang.Exception: shutdown Leader! reason: quorum Peer shutdown
15     at org.apache.zookeeper.server.quorum.Leader.shutdown(Leader.java:393)
16     at org.apache.zookeeper.server.quorum.QuorumPeer.shutdown(QuorumPeer.java:694)
17     at org.apache.zookeeper.server.quorum.QuorumPeerMain$1.run(QuorumPeerMain.java:126)
18 2011-06-21 10:09:59,733 - INFO [Thread-6:Leader$LearnerCnxAcceptor@243] - exception while shutting down acceptor: java.net.SocketException: Socket closed
19 2011-06-21 10:09:59,758 - INFO [ProcessThread:-1:PrepRequestProcessor@120] - PrepRequestProcessor exited loop!
20 2011-06-21 10:09:59,758 - INFO [CommitProcessor:2:CommitProcessor@150] - CommitProcessor exited loop!
21 2011-06-21 10:09:59,759 - INFO [Thread-2:FinalRequestProcessor@379] - shutdown of request processor complete
22 2011-06-21 10:10:00,000 - INFO [SessionTracker:SessionTrackerImpl@165] - SessionTrackerImpl exited loop!
```

```
"Thread-2" prio=10 tid=0x0810a400 nid=0x1695 in Object.wait() [0xac783000]
  java.lang.Thread.State: WAITING (on object monitor)
    at java.lang.Object.wait(Native Method)
      - waiting on <0xb804f5e8> (a org.apache.zookeeper.server.SyncRequestProcessor)
    at java.lang.Thread.join(Thread.java:1143)
      - locked <0xb804f5e8> (a org.apache.zookeeper.server.SyncRequestProcessor)
    at java.lang.Thread.join(Thread.java:1196)
    at org.apache.zookeeper.server.SyncRequestProcessor.shutdown(SyncRequestProcessor.java:171)
    at org.apache.zookeeper.server.quorum.ProposalRequestProcessor.shutdown(ProposalRequestProcessor.java:79)
    at org.apache.zookeeper.server.PrepareRequestProcessor.shutdown(PrepareRequestProcessor.java:513)
    at org.apache.zookeeper.server.ZooKeeperServer.shutdown(ZooKeeperServer.java:413)
    at org.apache.zookeeper.server.quorum.Leader.shutdown(Leader.java:411)
    at org.apache.zookeeper.server.quorum.QuorumPeer.shutdown(QuorumPeer.java:694)
    at org.apache.zookeeper.server.quorum.QuorumPeerMain$1.run(QuorumPeerMain.java:126)

"SyncThread:2" prio=10 tid=0xad7fd800 nid=0x4acb in Object.wait() [0xac9ba000]
  java.lang.Thread.State: WAITING (on object monitor)
    at java.lang.Object.wait(Native Method)
      - waiting on <0xb8030d00> (a org.apache.zookeeper.server.quorum.QuorumPeerMain$1)
    at java.lang.Thread.join(Thread.java:1143)
      - locked <0xb8030d00> (a org.apache.zookeeper.server.quorum.QuorumPeerMain$1)
    at java.lang.Thread.join(Thread.java:1196)
    at java.lang.ApplicationShutdownHooks.runHooks(ApplicationShutdownHooks.java:79)
    at java.lang.ApplicationShutdownHooks$1.run(ApplicationShutdownHooks.java:24)
    at java.lang.Shutdown.runHooks(Shutdown.java:79)
    at java.lang.Shutdown.sequence(Shutdown.java:123)
    at java.lang.Shutdown.exit(Shutdown.java:168)
      - locked <0xf01ff3b0> (a java.lang.Class for java.lang.Shutdown)
    at java.lang.Runtime.exit(Runtime.java:90)
    at java.lang.System.exit(System.java:904)
    at org.apache.zookeeper.server.SyncRequestProcessor.run(SyncRequestProcessor.java:149)
```

이슈번호 1109

애플리케이션 로그도 중요하지만
이 케이스의 경우 쓰레드 덤프도 중요

이슈번호 1513

ZooKeeper가 클라이언트의 쓰기는 정상적으로 받아서 transaction log에 저장했는데,
재시작하면서 그 transaction log를 자기 자신이 읽지 못해서 서버가 기동 실패하는 문제

SetDataTxn에는 클라이언트 데이터와 몇 가지 추가 필드가 포함됨.
서버는 데이터 수집 시 jute.maxbuffer 크기를 데이터에 적용하지만
디스크에서 읽을 때 데이터와 이러한 추가 필드의 크기가
jute.maxbuffer 검사값을 초과할 수 있다는 사실을 처리하지 못함

이슈번호 1513

0xffffffff

= 1,048,575 bytes

쓰기에도 이 제한을 통과해야 쓸 수 있음

그런데 제한을 통과한 후 들어오고 나서 트랜잭션 로그를 쓰기전에 메타데이터가 추가됨

저장에는 문제가 없었다가

재시동 후 트랜잭션로그를 읽어들이는 때 문제됨.

Unreasonable length 에러가 뜨며 종료됨

```
2012-07-18 14:28:12,555 - FATAL [main:QuorumPeer@400] - Unable to load database on disk
java.io.IOException: Unreasonable length = 1048583
at org.apache.jute.BinaryInputArchive.readBuffer(BinaryInputArchive.java:100)
at org.apache.zookeeper.server.persistence.Util.readTxnBytes(Util.java:232)
at org.apache.zookeeper.server.persistence.FileTxnLog$FileTxnIterator.next(FileTxnLog.java:602)
at org.apache.zookeeper.server.persistence.FileTxnLog$FileTxnIterator.init(FileTxnLog.java:529)
at org.apache.zookeeper.server.persistence.FileTxnLog$FileTxnIterator.<init>(FileTxnLog.java:504)
at org.apache.zookeeper.server.persistence.FileTxnLog.read(FileTxnLog.java:341)
at org.apache.zookeeper.server.persistence.FileTxnSnapLog.restore(FileTxnSnapLog.java:131)
at org.apache.zookeeper.server.ZKDatabase.loadDataBase(ZKDatabase.java:222)
at org.apache.zookeeper.server.quorum.QuorumPeer.start(QuorumPeer.java:398)
at org.apache.zookeeper.server.quorum.QuorumPeerMain.runFromConfig(QuorumPeerMain.java:143)
at org.apache.zookeeper.server.quorum.QuorumPeerMain.initializeAndRun(QuorumPeerMain.java:103)
at org.apache.zookeeper.server.quorum.QuorumPeerMain.main(QuorumPeerMain.java:76)
2012-07-18 14:28:12,555 - FATAL [main:QuorumPeerMain@87] - Unexpected exception, exiting abnormally
java.lang.RuntimeException: Unable to run quorum server
at org.apache.zookeeper.server.quorum.QuorumPeer.start(QuorumPeer.java:401)
at org.apache.zookeeper.server.quorum.QuorumPeerMain.runFromConfig(QuorumPeerMain.java:143)
at org.apache.zookeeper.server.quorum.QuorumPeerMain.initializeAndRun(QuorumPeerMain.java:103)
at org.apache.zookeeper.server.quorum.QuorumPeerMain.main(QuorumPeerMain.java:76)
Caused by: java.io.IOException: Unreasonable length = 1048583
at org.apache.jute.BinaryInputArchive.readBuffer(BinaryInputArchive.java:100)
at org.apache.zookeeper.server.persistence.Util.readTxnBytes(Util.java:232)
at org.apache.zookeeper.server.persistence.FileTxnLog$FileTxnIterator.next(FileTxnLog.java:602)
at org.apache.zookeeper.server.persistence.FileTxnLog$FileTxnIterator.init(FileTxnLog.java:529)
at org.apache.zookeeper.server.persistence.FileTxnLog$FileTxnIterator.<init>(FileTxnLog.java:504)
at org.apache.zookeeper.server.persistence.FileTxnLog.read(FileTxnLog.java:341)
at org.apache.zookeeper.server.persistence.FileTxnSnapLog.restore(FileTxnSnapLog.java:131)
at org.apache.zookeeper.server.ZKDatabase.loadDataBase(ZKDatabase.java:222)
at org.apache.zookeeper.server.quorum.QuorumPeer.start(QuorumPeer.java:398)
... 3 more
```

이슈번호 1560

Java 클라이언트를 사용하여 0.5MB의 데이터로 노드를 생성해 보세요.
테스트는 서버 응답을 기다리는 동안 멈춥니다

문제 1 — 아직 다 안 보냈는데 Queue에서 패킷을 제거함

문제 2 — ByteBuffer를 계속 새로 만들어버림

문제 3 — XID까지 패킷 조각마다 증가

```
80 EndOfStreamException: Unable to read additional data from client sessionid 0x0, likely client has closed socket
81     at org.apache.zookeeper.server.NIOServerCnxn.doIO(NIOServerCnxn.java:220)
82     at org.apache.zookeeper.server.NIOServerCnxnFactory.run(NIOServerCnxnFactory.java:208)
83     at java.lang.Thread.run(Thread.java:704)
84 2026-09-16 02:32:47,398 [myid:] - INFO [NIOServerCxn.Factory:0.0.0.0/0.0.0.0:2181:NIOServerCnxn@1001] - Closed socket connection for client /172.17.0.1:54596 (no session established for client)
85 2026-09-16 02:33:21,823 [myid:] - INFO [NIOServerCxn.Factory:0.0.0.0/0.0.0.0:2181:NIOServerCnxnFactory@197] - Accepted socket connection from /172.17.0.1:52314
86 2026-09-16 02:33:21,831 [myid:] - INFO [NIOServerCxn.Factory:0.0.0.0/0.0.0.0:2181:ZooKeeperServer@839] - Client attempting to establish new session at /172.17.0.1:52314
87 2026-09-16 02:33:21,836 [myid:] - INFO [SyncThread:0:FileTxnLog@199] - Creating new log file: log.1
88 2026-09-16 02:33:21,873 [myid:] - INFO [SyncThread:0:ZooKeeperServer@595] - Established session 0xffa0a80f13070000 with negotiated timeout 10000 for client /172.17.0.1:52314
89 2026-09-16 02:33:28,608 [myid:] - WARN [NIOServerCxn.Factory:0.0.0.0/0.0.0.0:2181:NIOServerCnxn@349] - caught end of stream exception
90 EndOfStreamException: Unable to read additional data from client sessionid 0xffa0a80f13070000, likely client has closed socket
91     at org.apache.zookeeper.server.NIOServerCnxn.doIO(NIOServerCnxn.java:220)
92     at org.apache.zookeeper.server.NIOServerCnxnFactory.run(NIOServerCnxnFactory.java:208)
93     at java.lang.Thread.run(Thread.java:704)
94 2026-09-16 02:33:28,611 [myid:] - INFO [NIOServerCxn.Factory:0.0.0.0/0.0.0.0:2181:NIOServerCnxn@1001] - Closed socket connection for client /172.17.0.1:52314 which had sessionid 0xffa0a80f13070000
95 2026-09-16 02:33:30,585 [myid:] - INFO [NIOServerCxn.Factory:0.0.0.0/0.0.0.0:2181:NIOServerCnxnFactory@197] - Accepted socket connection from /172.17.0.1:54196
96 2026-09-16 02:33:30,587 [myid:] - INFO [NIOServerCxn.Factory:0.0.0.0/0.0.0.0:2181:ZooKeeperServer@832] - Client attempting to renew session 0xffa0a80f13070000 at /172.17.0.1:54196
97 2026-09-16 02:33:30,590 [myid:] - INFO [NIOServerCxn.Factory:0.0.0.0/0.0.0.0:2181:ZooKeeperServer@595] - Established session 0xffa0a80f13070000 with negotiated timeout 10000 for client /172.17.0.1:54196
98 2026-09-16 02:33:52,841 [myid:] - WARN [NIOServerCxn.Factory:0.0.0.0/0.0.0.0:2181:NIOServerCnxn@349] - caught end of stream exception
99 EndOfStreamException: Unable to read additional data from client sessionid 0xffa0a80f13070000, likely client has closed socket
100     at org.apache.zookeeper.server.NIOServerCnxn.doIO(NIOServerCnxn.java:220)
101     at org.apache.zookeeper.server.NIOServerCnxnFactory.run(NIOServerCnxnFactory.java:208)
102     at java.lang.Thread.run(Thread.java:704)
103 2026-09-16 02:33:52,844 [myid:] - INFO [NIOServerCxn.Factory:0.0.0.0/0.0.0.0:2181:NIOServerCnxn@1001] - Closed socket connection for client /172.17.0.1:54196 which had sessionid 0xffa0a80f13070000
104 2026-09-16 02:34:02,001 [myid:] - INFO [SessionTracker:ZooKeeperServer@325] - Expiring session 0xffa0a80f13070000, timeout of 10000ms exceeded
105 2026-09-16 02:34:02,002 [myid:] - INFO [ProcessThread(sid:0 cport:-1)::PrepRequestProcessor@476] - Processed session termination for sessionid: 0xffa0a80f13070000
```

```
"main-EventThread" daemon prio=10 tid=0x0000789a9c1f4000 nid=0x3025c waiting on condition [0x0000789a913fe000]
  java.lang.Thread.State: WAITING (parking)
    at sun.misc.Unsafe.park(Native Method)
      - parking to wait for <0x00000007accd4888> (a java.util.concurrent.locks.AbstractQueuedSynchronizer$ConditionObject)
    at java.util.concurrent.locks.LockSupport.park(LockSupport.java:186)
    at java.util.concurrent.locks.AbstractQueuedSynchronizer$ConditionObject.await(AbstractQueuedSynchronizer.java:2043)
    at java.util.concurrent.LinkedBlockingQueue.take(LinkedBlockingQueue.java:386)
    at org.apache.zookeeper.ClientCnxn$EventThread.run(ClientCnxn.java:491)
```

Locked ownable synchronizers:

- None

```
"main-SendThread(localhost:12182)" daemon prio=10 tid=0x0000789a9c1f3000 nid=0x3025b runnable [0x0000789a917fe000]
  java.lang.Thread.State: RUNNABLE
    at sun.nio.ch.EPollArrayWrapper.epollWait(Native Method)
    at sun.nio.ch.EPollArrayWrapper.poll(EPollArrayWrapper.java:228)
    at sun.nio.ch.EPollSelectorImpl.doSelect(EPollSelectorImpl.java:83)
    at sun.nio.ch.SelectorImpl.lockAndDoSelect(SelectorImpl.java:87)
      - locked <0x00000007acca40d0> (a sun.nio.ch.Util$1)
      - locked <0x00000007acca4070> (a java.util.Collections$UnmodifiableSet)
      - locked <0x00000007acca39c8> (a sun.nio.ch.EPollSelectorImpl)
    at sun.nio.ch.SelectorImpl.select(SelectorImpl.java:98)
    at org.apache.zookeeper.ClientCnxnSocketNIO.doTransport(ClientCnxnSocketNIO.java:336)
    at org.apache.zookeeper.ClientCnxn$SendThread.run(ClientCnxn.java:1068)
```

Locked ownable synchronizers:

- None

```
"main" prio=5 tid=7f9bed000800 nid=0x10c382000 in Object.wait() [10c380000]  
  java.lang.Thread.State: WAITING (on object monitor)  
    at java.lang.Object.wait(Native Method)  
      - waiting on <7dc1a15c0> (a org.apache.zookeeper.ClientCnxn$Packet)  
    at java.lang.Object.wait(Object.java:485)  
    at org.apache.zookeeper.ClientCnxn.submitRequest(ClientCnxn.java:1309)  
      - locked <7dc1a15c0> (a org.apache.zookeeper.ClientCnxn$Packet)  
    at org.apache.zookeeper.ZooKeeper.create(ZooKeeper.java:781)  
    at org.apache.zookeeper.test.ClientTest.testLargeNodeData(ClientTest.java:531)
```

이슈번호 1560

이 케이스 또한 쓰레드 덤프가 중요해보임

JVM 쓰레드 덤프

스레드 덤프는 실행 중인 자바 애플리케이션의 모든 스레드 상태를 출력한 스냅샷이다.

서버에서 응답 지연, 무한 대기, CPU 사용률 급증, 데드락 등이 발생했을 때, 그 원인을 파악하기 위한 주요 도구로 사용된다.

스레드 덤프를 분석하면 다음을 확인할 수 있다:

- 현재 어떤 스레드가 실행 중인지
- 어떤 스레드가 락을 점유하고 있는지,
- 어떤 스레드가 락을 기다리고 있는지
- 데드락 여부 및 락 경합 병목 구간